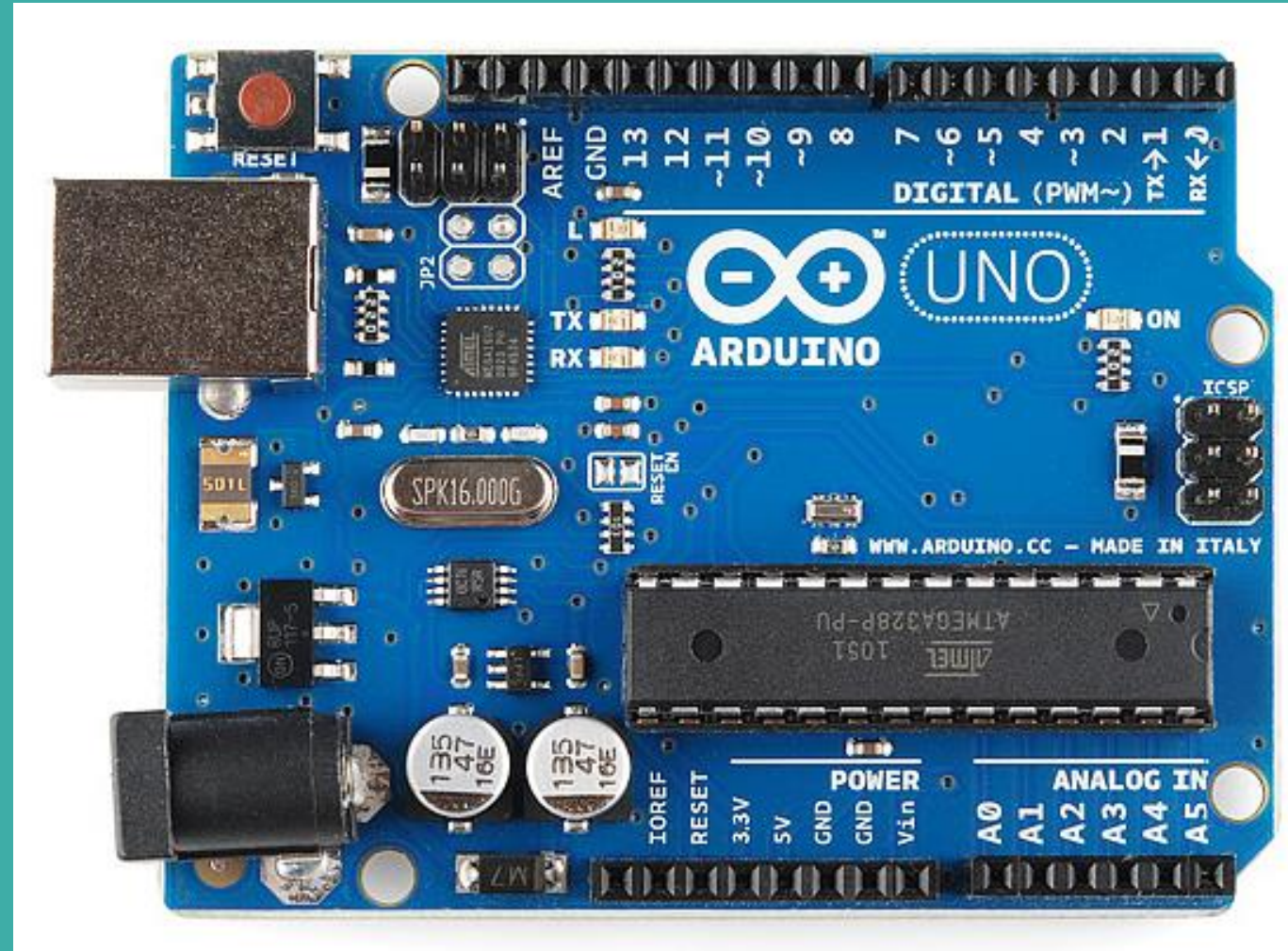


Android-Arduino Integration

RECAP

Arduino Programming



Mobile programming

The Easy Way to Create Your Own App



MIT App Inventor



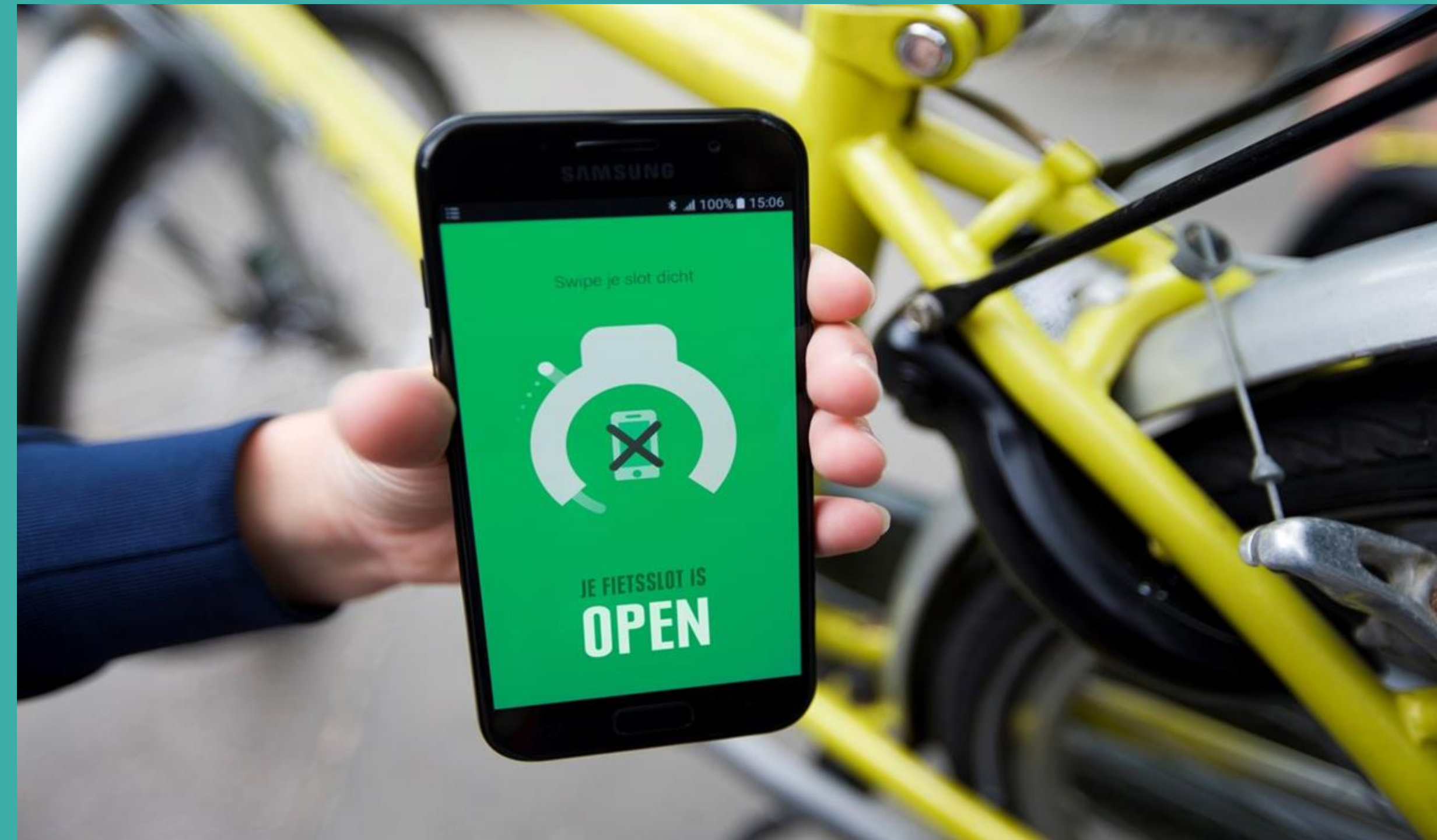
Arduino + Android

All powers of custom hardware and custom software combined! Since ~2010

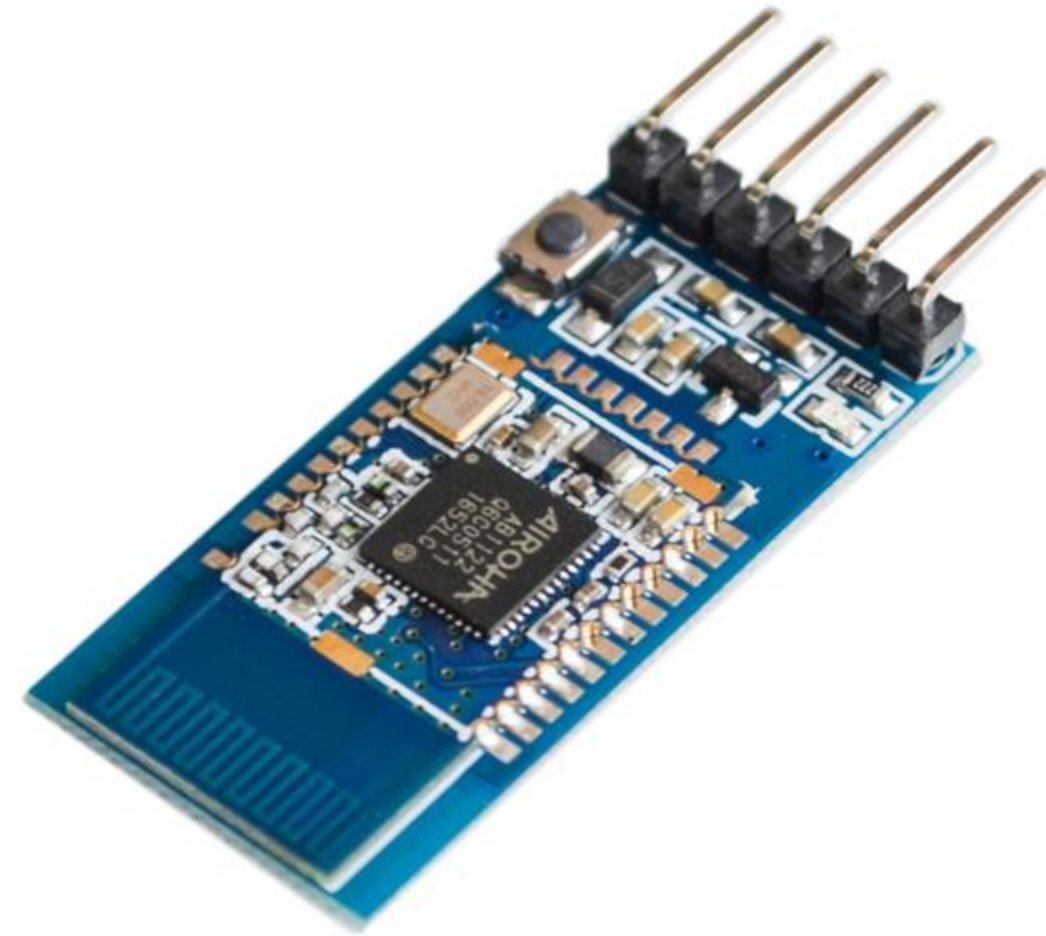
You see it in everything Smart!



 **Noon Smart Lighting**



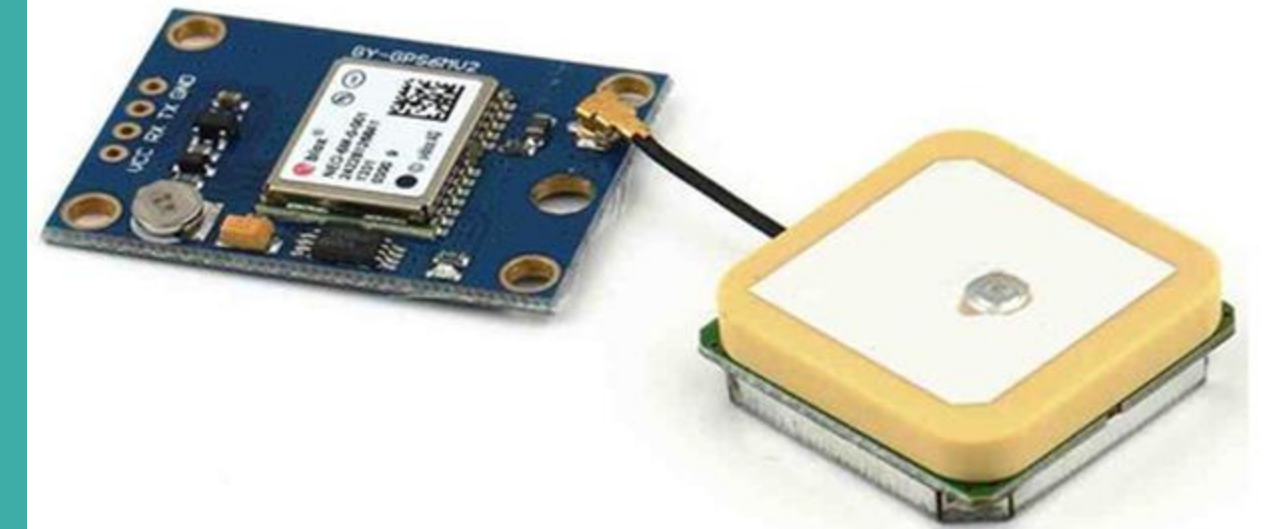
Wireless Ways to Communicate with Arduino



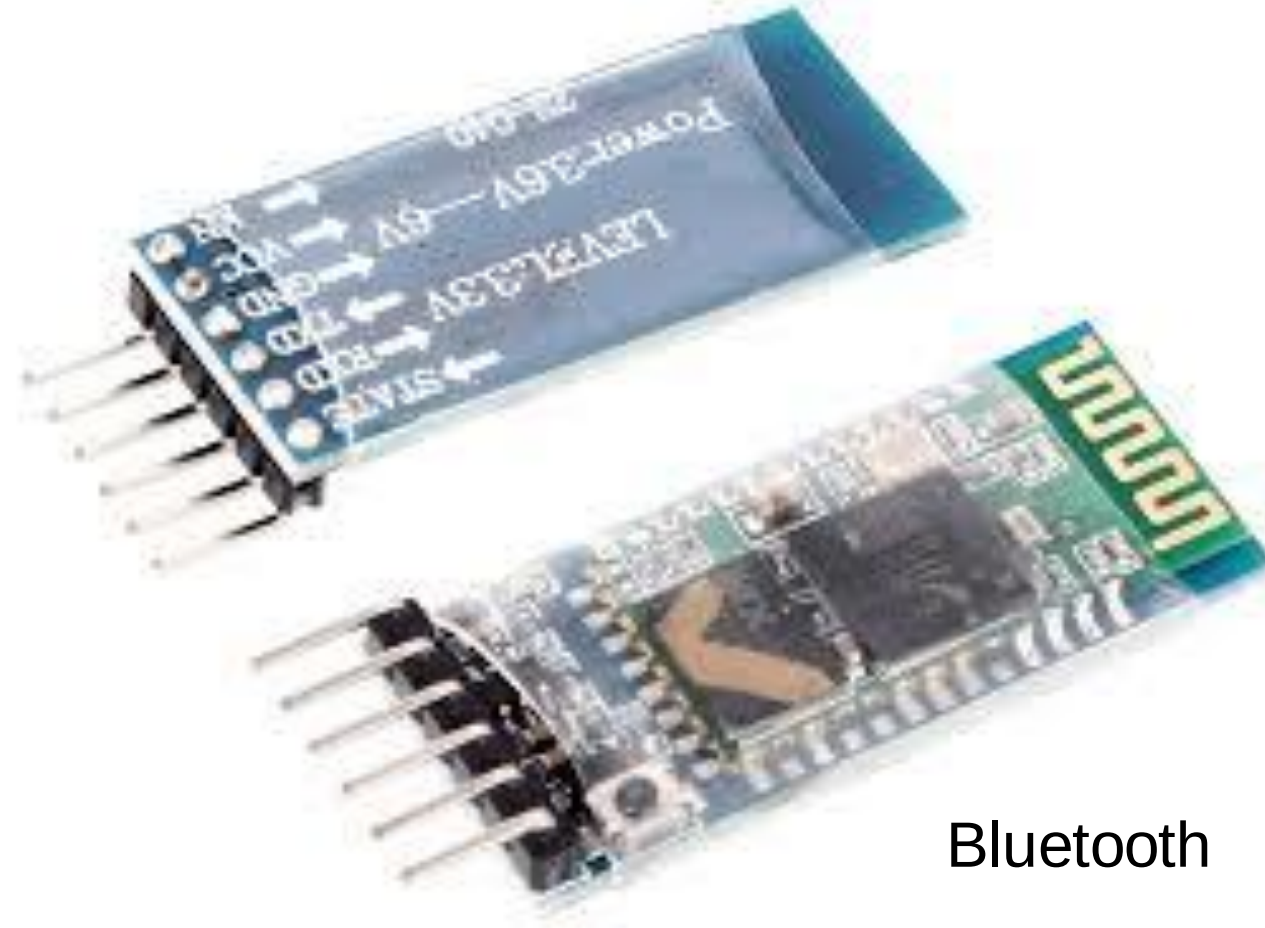
BLE 4.0



Wi-Fi



PS
G



Bluetooth



Zigbee



GSM

Bluetooth

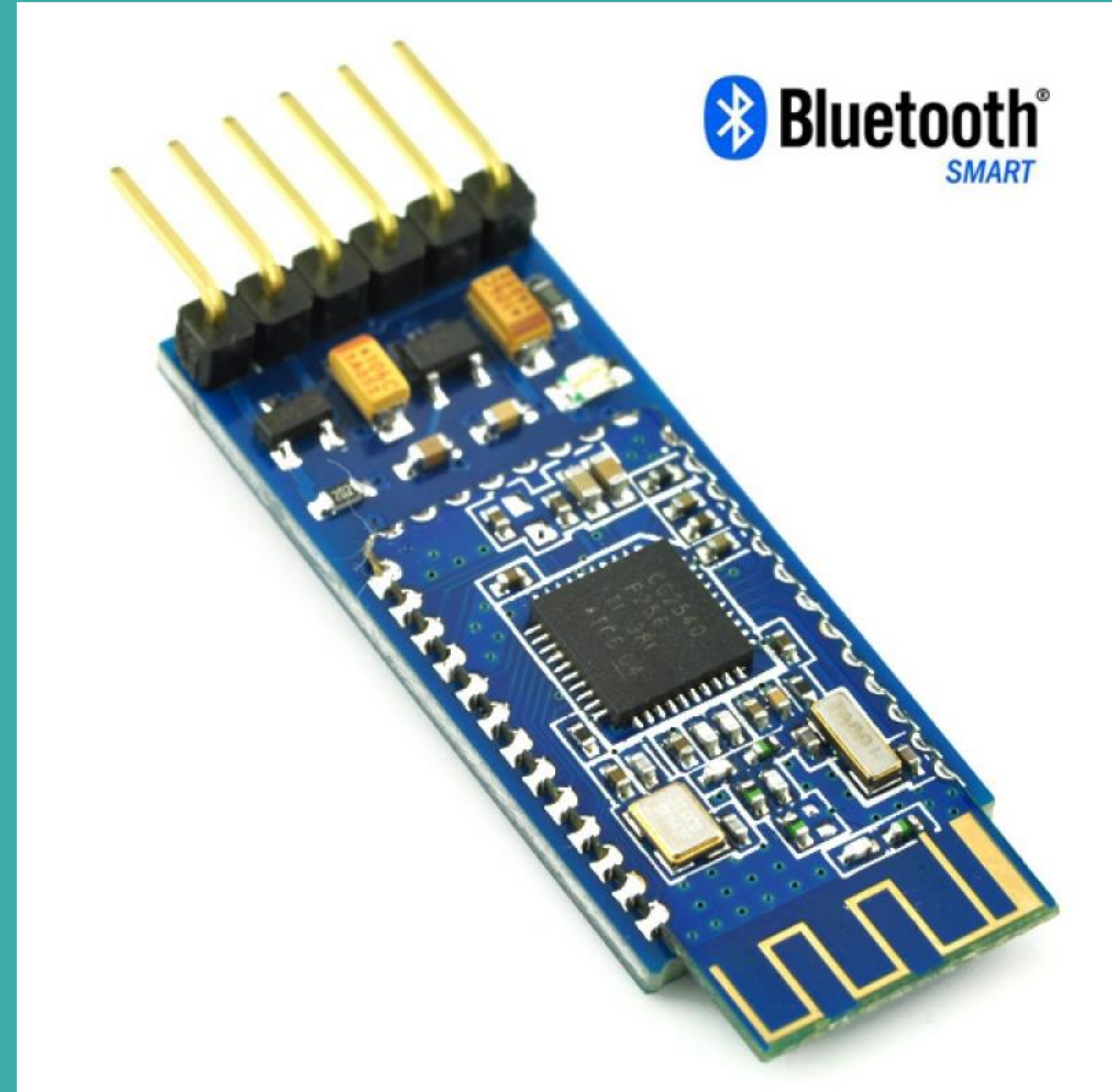
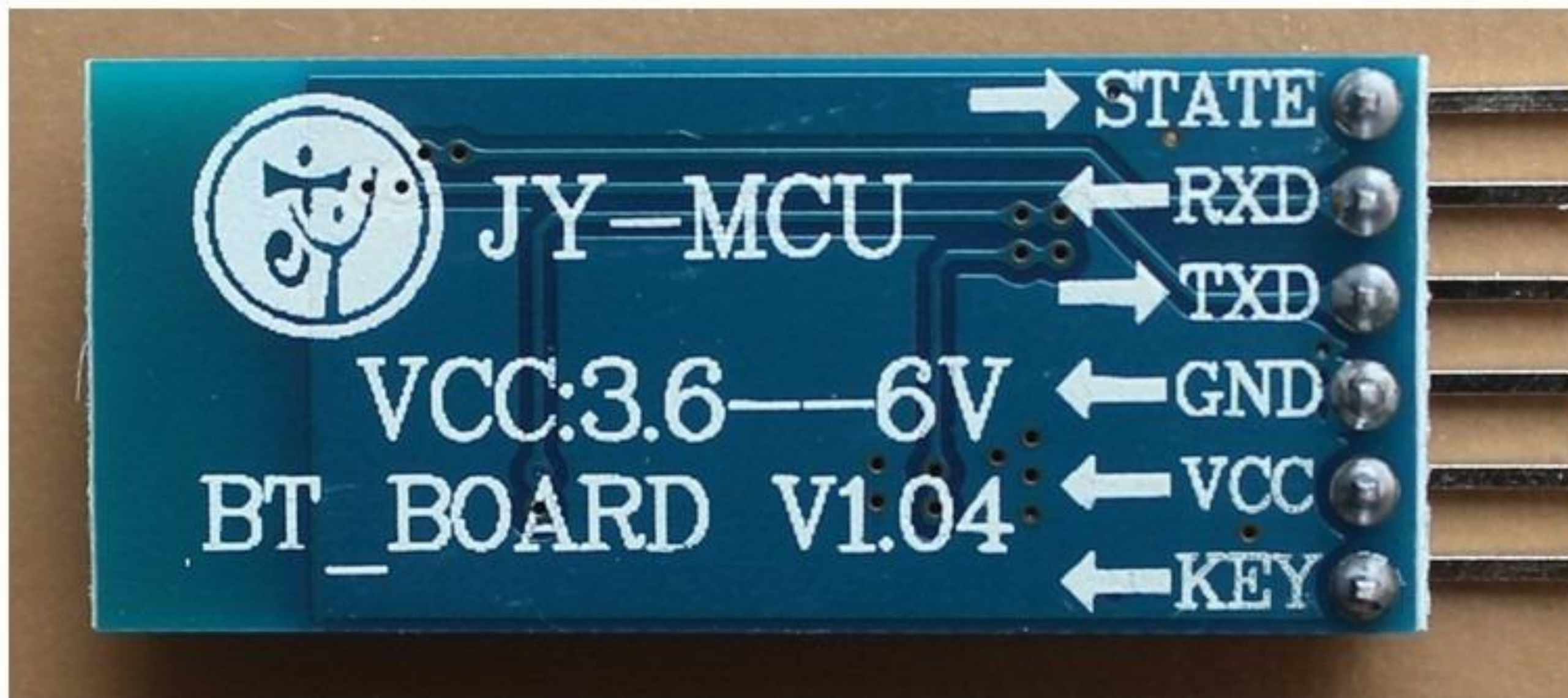
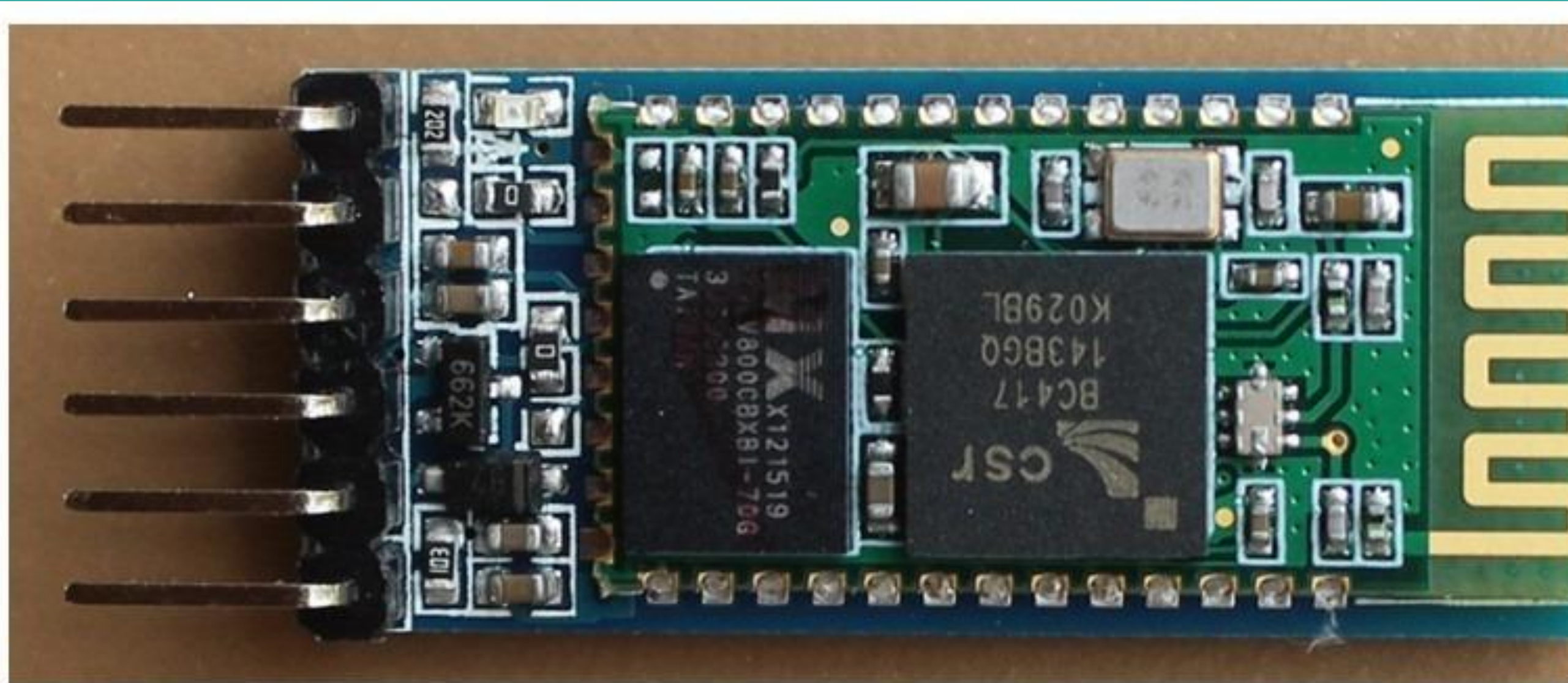


It is a simple way for cell phones, printers, PCs, digital cameras, and other gadgets to link together over relatively short distances using wireless (radio wave) technology.

Range: 10m or 30ft



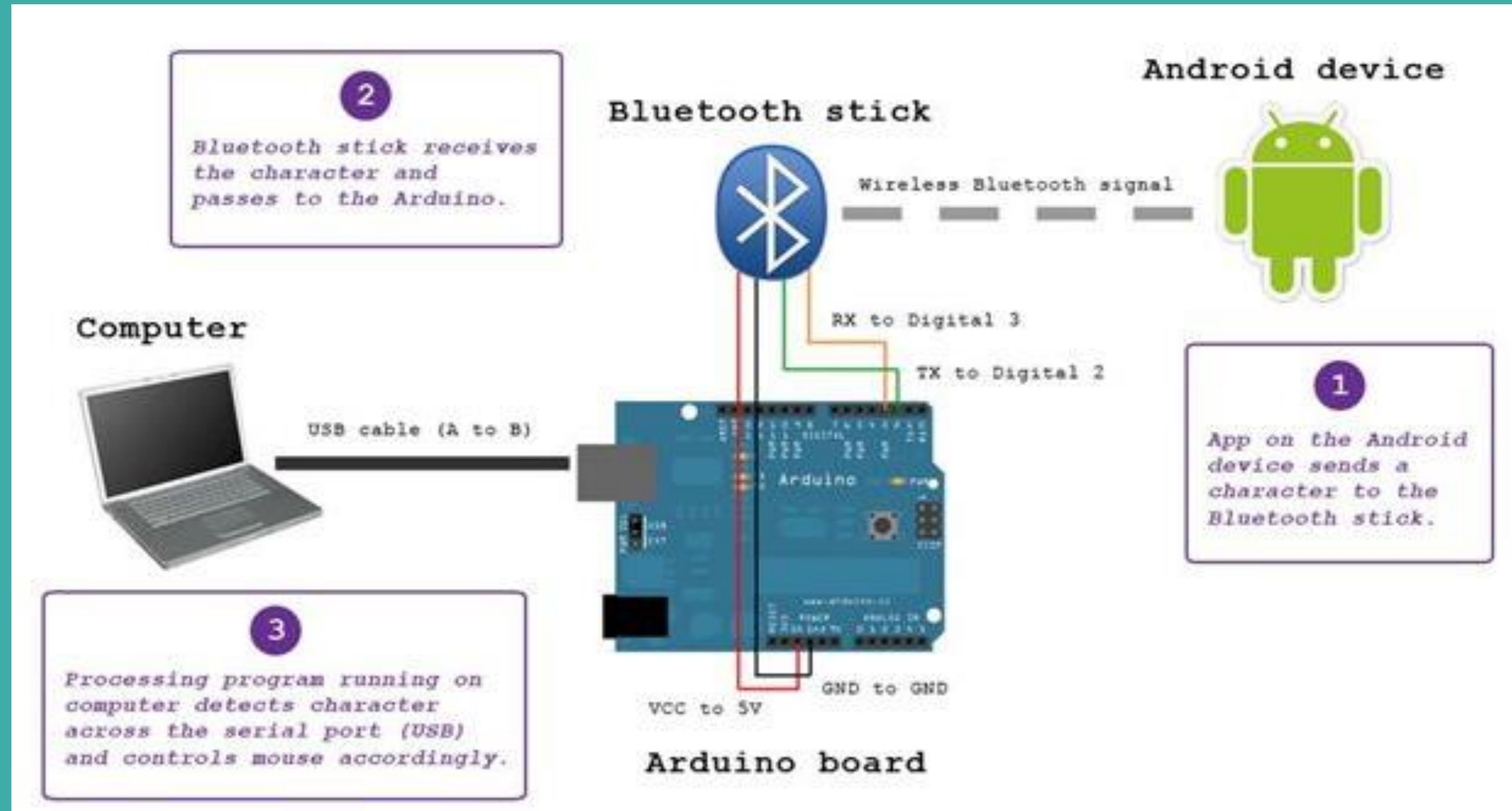
Bluetooth Module Package and their Pins



Activity 1

Control LED on Arduino uno with App developed in MIT
APP Inventor using Bluetooth module

Signal Flow diagram



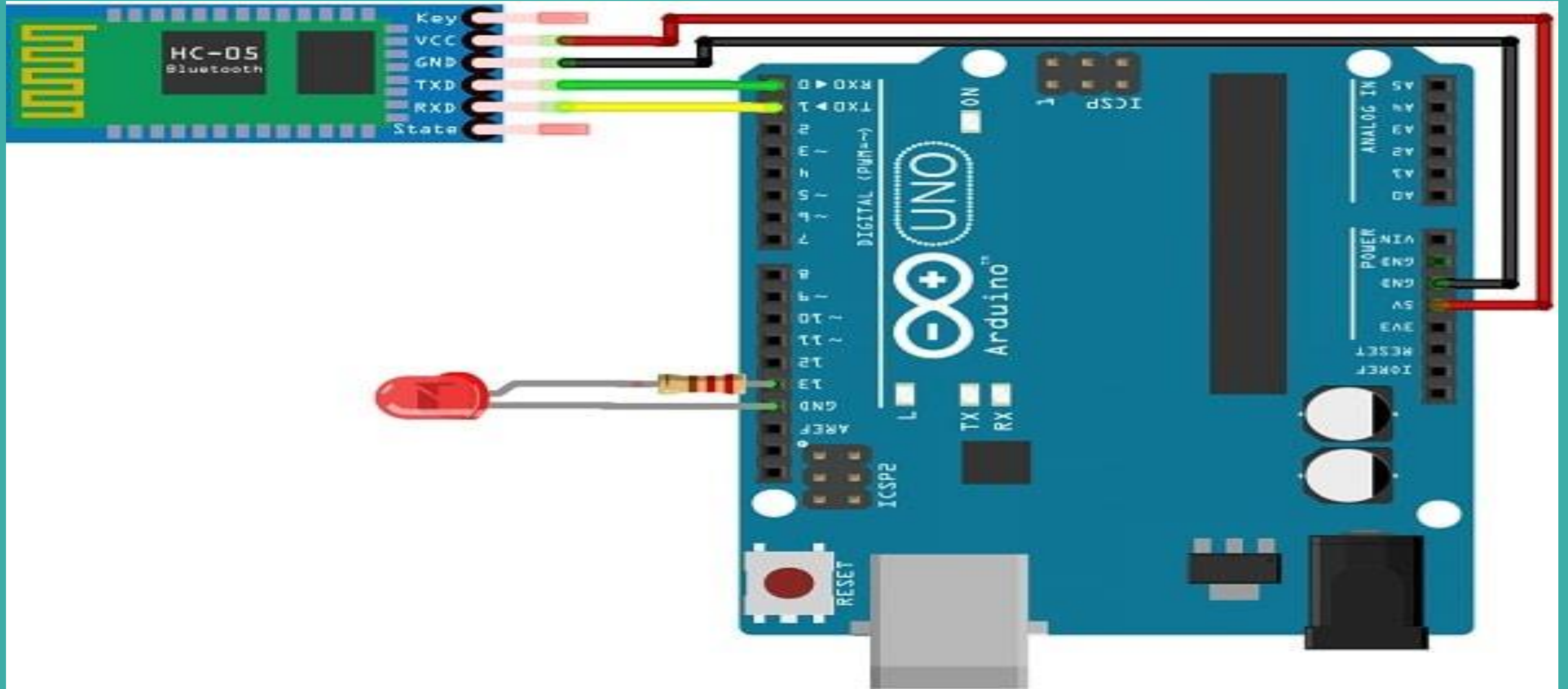
What you will need:

- Computer with Internet access
- Android smartphone or tablet
- Arduino
- HC-05 Bluetooth module
- Free account at MIT App Inventor website
- LED , Breadboard , Wires , Potentiometer



Hc-05 Bluetooth
SSID: HC05
Passkey: 1234

Circuit Diagram



Code for the Arduino to send and receive data from BLuetooth and Android app

```
#include <SoftwareSerial.h>
SoftwareSerial mySerial(10,11);           // 10- Rx , 11 - Tx
int ledpin=13;
int Data;

void setup()
{
  mySerial.begin(9600);
  pinMode(ledpin,OUTPUT);
}
void loop()
{

  if (mySerial.available())
  {
    Data=mySerial.read();
    if(Data=='1')
    {
      digitalWrite(ledpin,HIGH);
      mySerial.println("LED On! ");
    }
    else if (Data=='0')
    {
      digitalWrite(ledpin,LOW);
      mySerial.println("LED Off! ");
    }
  }

}
```


App Inventor

<https://goo.gl/eVYvh3>

Expected Output



Activity 2

Read values from sensor connected to Arduino uno and display them on the Android App developed in MIT APP Inventor using Bluetooth module

References

<http://www.instructables.com/id/Remotely-Control-LED-using-HC-05-Bluetooth-Arduino/>

<http://appinventor.mit.edu/extensions/>

http://iot.appinventor.mit.edu/assets/howtos/MIT_App_Inventor_Basic_Connection.pdf

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